

AI-Powered Sentiment Tracking and Support System for Emotional Well-Being

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A THESIS SUBMITTED IN FULFILLMENT OF THE REQUIREMENTS

FOR THE DEGREE OF

Bachelor of Science

IN

Computer Science



Department of Computer Science and Technology

Wenzhou-Kean University

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© June 2026

Bachelor of Science (2026)
Computer Science and Technology

Wenzhou-Kean University
Wenzhou, China

TITLE:	AI-Powered Sentiment Tracking and Support System for Emotional Well-Being
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NUMBER OF PAGES:	49

Abstract

The goal of this capstone project is to conceptualize, execute, and assess an emotion-tracking device that protects the individual's right to privacy using natural language processing (NLP) techniques with a transformer architecture on a person's journaling text. The system is intended for university students and young professionals who frequently experience academic, social, and occupational stress yet may hesitate to use cloud-based mental health tools because of privacy concerns. The primary goal of this project is to create a system that allows users to reflect on their emotional states in a private and accessible way while still receiving meaningful AI-assisted feedback.

The application is delivered as a web-based system consisting of a Python/Flask back end and a Vue.js + Chart.js front end. Users enter short diary-style text, and the system returns probability scores for multiple emotion categories together with a visualization of emotional trends over time. To build the emotion classifier, multiple publicly available emotion datasets were integrated, cleaned, and harmonized into a unified multi-label learning dataset. The implementation explores fine-tuned transformer models, including DistilBERT and RoBERTa, together with modern embedding-based

representations in order to balance predictive quality and computational efficiency.

The local-first design principle was employed throughout the project. The REST API exposed by the system allows the user to store their emotion history locally, which reduces reliance on remote cloud computation and also protects sensitive personal text from being sent to a remote location. The emotion classifier was evaluated quantitatively via precision, recall, F1 score, and multi-label accuracy metrics. Additionally, the System Usability Scale and a short user satisfaction questionnaire were used to evaluate the system qualitatively.

This project shows that modern NLP models can be integrated into a lightweight application that is aware of privacy and that supports emotional self-reflection. The work also identifies practical constraints and future opportunities for personalization; resource-efficient deployment; interpretability; and ethically responsible emotion-aware user interfaces.

Keywords: Emotion Recognition, Transformers, Privacy-Preserving AI, Journaling Analysis, Natural Language Processing

Acknowledgements

I would like to acknowledge and thank Dr. Rashid Sangi, my adviser for the guidance and valuable feedback provided throughout this capstone project. Their assistance helped me enhance my research focus, refine my methods, and improve the final product – both technically and academically.

I wish to recognize my class mates and friends, who provided me with their thoughts and critiques as I developed and evaluated this project. Their constructive comments made it possible to increase the usability and understandability of the system, through participant testing and group discussions.

Lastly, I would like to acknowledge the open-source community and the researchers whose publicly available NLP models, datasets, and development tools made this project possible.

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Definition of Acronyms

Acronym	Definition
LightGBM	Light Gradient Boosting Machine
MLP	Multilayer Perceptron
CNN	Convolutional Neural Network
LSTM	Long Short-Term Memory
GNN	Graph Neural Network
LIME	Local Interpretable Model-agnostic Explanations
HTML	HyperText Markup Language
LLM	Large Language Model
AI	Artificial Intelligence
GPU	Graphics Processing Unit
ROC	Receiver Operating Characteristic
PR	Precision-Recall
AUC	Area Under Curve
AP	Average Precision
SHAP	SHapley Additive exPlanations
SUS	System Usability Scale
BCE	Binary Cross-Entropy

Definition of Notations

Symbol	Meaning
n	Number of samples or elements
x	An input variable or data point
μ	Mean (average)
σ	Standard deviation
$\sum$	Summation operator
K	Number of emotion labels in the multi-label classification setting
z_k	Logit for label k before sigmoid activation
p_k	Predicted probability for label k after sigmoid activation
τ	Decision threshold used to determine whether a label is displayed

Chapter 1

Introduction

The importance of emotional and sentiment analysis is growing in the fields of AI, natural language processing (NLP), and user-centred computing. This thesis presents an investigation into how new transformers can aid in forming private self-reflection using a journaling application versus just regular text classification. This project has been initiated due to a practical design challenge with respect to providing a satisfactory solution: to create a platform that provides a lightweight solution that has enough accuracy to classify multiple emotions from free-form data with user control of the emotional content.

1.1 Introduction to Emotion Analysis and Sentiment Analysis

The last few years have seen a major shift in the intersection of AI and psychology through the growth of affective computing, which studies how computational systems

can detect, interpret, and respond to human emotions. Recent examinations indicate an increase both in numbers of young adults in our society and also their struggles with Mental Health due to Stressors, Uncertainty, Burnout, & Social Isolation. In addition to this need - we have also seen an increase in demand from these young adults for more accessible digital tools that help promote emotional self-awareness.

In the past, the main application of NLP has been through sentiment analysis, where, in a commercial setting, sentiment is typically aggregated into positive, negative and neutral polarity labels. These categories, while useful for brand tracking, assessing customer reviews, and mining opinions provide an incomplete view of a person's emotional state during reflection. Most people's emotions can not be classified as only positive or negative; a person's experience could be reflected as relief, sadness, uncertainty, gratitude and/or fatigue in a diary entry. A more nuanced emotion identification framework should be applied rather than rely solely on traditional sentiment polarity [2, 5].

An example of this could be a journal entry where someone writes "I finally finished my thesis, but now I feel empty because it is over." The positive feeling of accomplishment and the empty feeling from finishing the thesis are both emotions present in this entry. A typical sentiment classifier may identify the word "finished" as positive and fail to capture much about the sense of fatigue or loss associated with completing a significant task. Thus, the limitations of traditional polarity-based sentiment analysis need to be considered when examining the effectiveness of using sentiment analysis for self-reflection and emotional wellness assessments.

Whereas feelings (aka Emotion Recognition) can be identified based upon a binary approach - either positive or negative - identifying emotional states (e.g., happy, upset, afraid, sad, amazed, loving, optimistic, disappointed, etc.) requires a distinct third-level of classification to include all of the emotions we experience as human beings. This distinction becomes particularly important when using journaling type systems for recording short unstructured and/or context (situationally dependent) focused textual content, as the manner in which emotion is established through language has to do with more than just the words themselves (e.g., words provide meanings relative to context, contrast/negation, and failure to provide an emotional context). Analyses of Data using Modern Transformer Architecture Models (e.g., BERT, RoBERTa & T5) are highly relevant because current transformer-based approaches are better at modeling information relative to context than other past/natural language processing techniques (e.g., Word2vec) which do not utilize more advanced methods of modeling contextual information than those previously noted [1,3,4].

In addition, emotional writing has long been associated with expressive writing and reflective journaling practices. Prior work in psychology suggests that structured reflection through writing can support emotional processing and self-understanding [6]. Yet many journaling tools provide only storage or visualization features and do not offer intelligent analysis. Many AI tools for analyzing language are very powerful for analyzing text; however, many of them operate in a cloud-based processing model where you may not want to have your input text exposed to the is cloud because of its highly personal nature or of a more psychological, sensitive nature.

As a result, this project will also propose that meaningful emotional support tools need to be developed using a multi-label emotion recognition model rather than just a sentiment polarity-based model and also take privacy into consideration as an important aspect of the system itself. The goal is not clinical diagnosis, but emotional awareness: helping users observe patterns in their own writing over time in a way that is transparent, non-intrusive, and privacy-conscious.

1.1.1 Importance of Privacy in Artificial Intelligence Mental Health Systems

The rise of artificial intelligence (AI) systems in our daily lives has raised serious questions regarding privacy, independence and trust; especially when applied to health and personal behaviours. Current cloud-based AI services typically require users to send raw text to the service’s remote processing servers. Although this provides easy access to powerful computer resources, they can also come with potential privacy issues such as servers exposing user data to outside parties, storage breaches, models logging information on users or nuisance uses of any of that data.

This is especially true in the case of emotional journaling applications e.g. journals or diaries that can record your thoughts or feelings about relationships with friends, family issues, school anxiety, etc. Even after removing the obvious identifiers, there can still be enough detail within any handwritten form of text to identify the writer or their situation. Therefore, if users feel that their text will be kept or processed beyond their controls, they may decide not to use a digital reflection tool in a completely honest

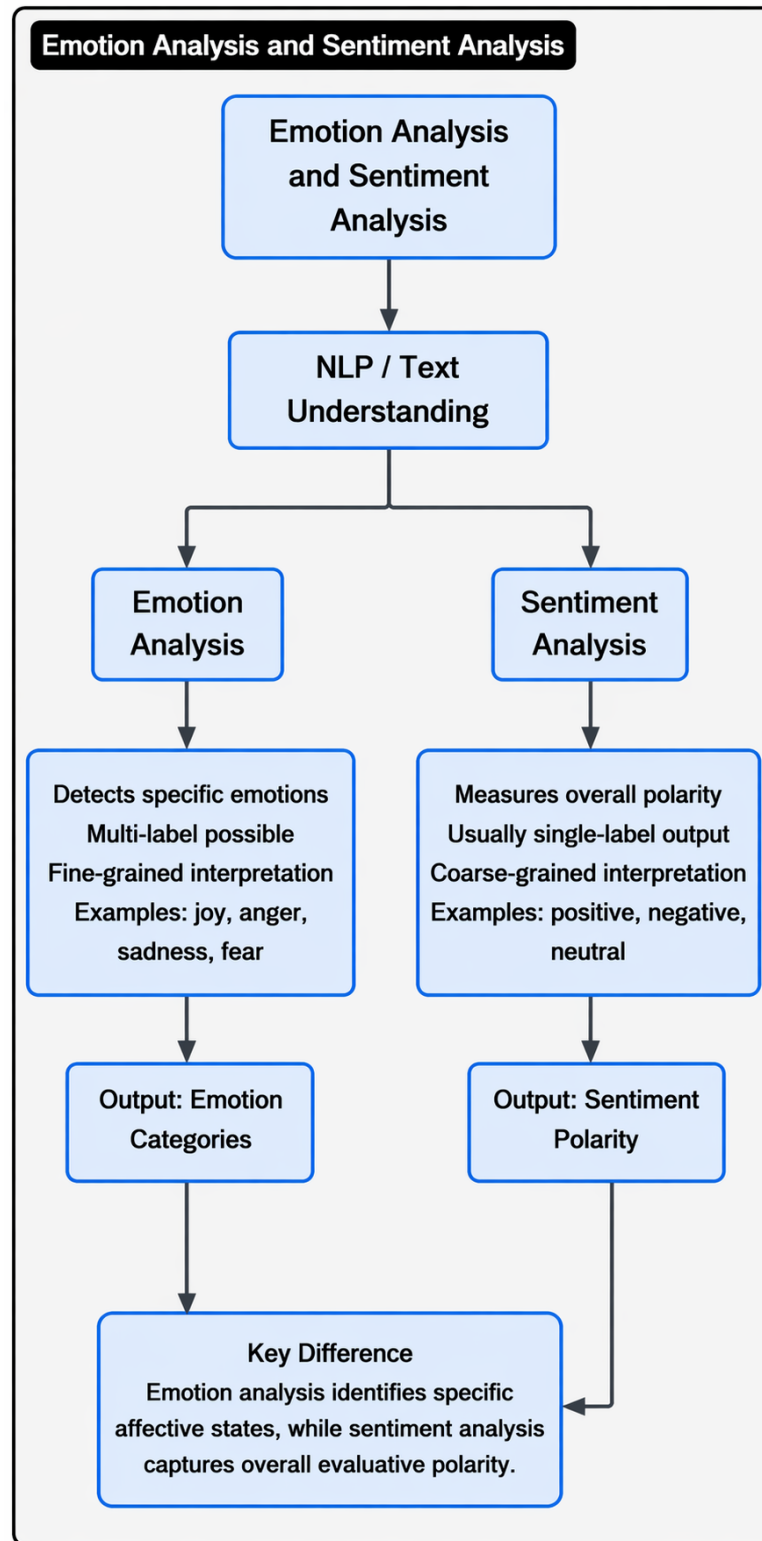


Figure 1.1: Emotion Analysis and Sentiment Analysis Overview

manner as a result.

Therefore, this thesis considers privacy as an architectural requirement rather than just an optional feature. The system design emphasises local-first inference, minimal data retention, the user decides what history will be retained, and a transparent view of system limitations. Instead of presuming that users will exchange privacy for convenience, the system assumes that the socially responsible way to design an AI product is to provide a product that is both easy to use, while protecting users' rights over their data. This design consideration aligns with current trends associated with edge AI and local deployment, as lightweight transformer models enable increasingly feasible on-device or controlled local inference [7–9].

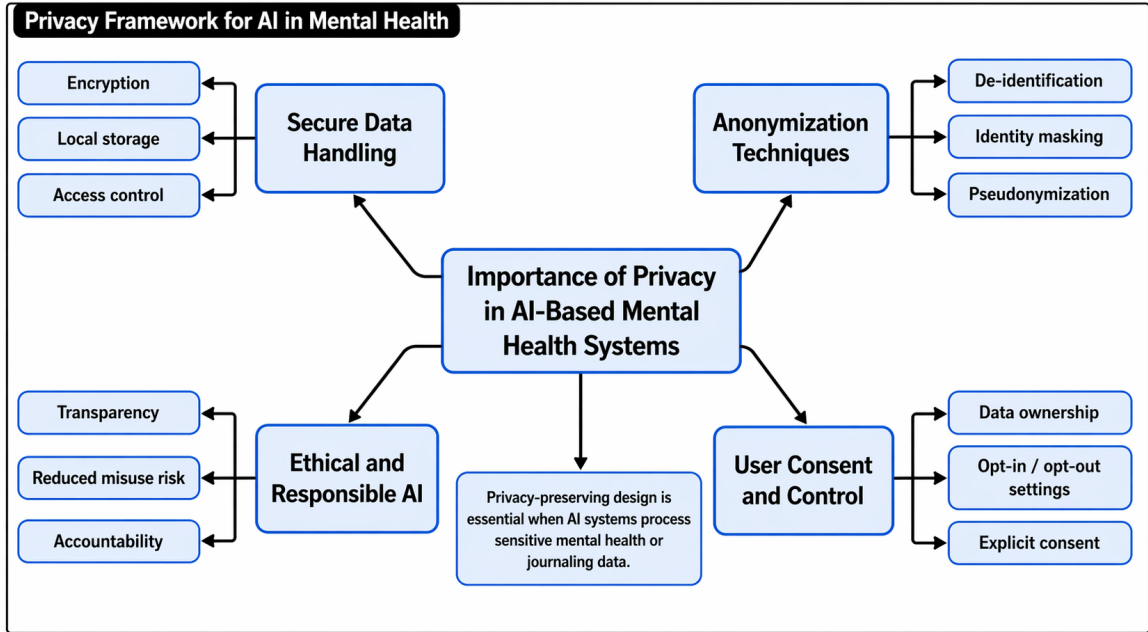


Figure 1.2: Privacy Considerations in AI-Based Mental Health Systems

1.2 Motivation and Problem Statement

1.2.1 Motivation

This research is motivated by two related concerns: the need for better support for emotional self-awareness among younger users and the privacy challenges associated with digital mental health technologies.

University students and early-career professionals often experience overlapping sources of stress, including academic workload, financial pressure, social uncertainty, and career anxiety. Oftentimes the gradual added strain from emotional sources develops over time versus all at once. An individual may not realize they have early symptoms of burnout, frustration or an unstable emotional state until such symptoms increase. The practice of Journaling is prevalent in its recognition as a productive tool for self-reflection to help one to put words to their internal experiences and to be able to review those experiences in the future. However, journaling becomes more effective when users receive structured feedback that helps them identify trends, recurring emotions, and changes in intensity.

At the same time, many existing digital support tools depend heavily on cloud infrastructure. Users may enter text or data into an online service that is sensitive (e.g. a personal journal). This data could either be sent, processed, or kept in a way that the user may not fully understand. The fact that a service claims to be secure does not guarantee the user will trust the service - the mere fear of being exposed

could create distrust between them and the provider. Building trust is an integral part of successfully providing an emotionally sensitive application.

Therefore this project was created with the intent of finding ways in which to use privacy-supporting AI techniques to improve the trustworthiness of using journaling technologies, while at the same time providing these users with a practical and ethical way of tracking their feelings over a long period of time. This thesis does not focus on creating another large, cloud-based emotion recognition solution, but rather how to support users of emotion recognition services through a user owned, local-first option that will provide similar insight into how they feel without requiring them to provide their own private/secure text data to others.

1.2.2 Problem Statement

Developing such a system is challenging because it requires balancing technical performance, usability, and privacy constraints. This thesis focuses on three concrete gaps.

Performance–Resource Trade-off: State-of-the-art transformer models often deliver strong accuracy, but many of them are too computationally expensive for lightweight or local deployment. There is a practical need for architectures that maintain acceptable multi-label classification performance while remaining deployable in constrained environments.

Multi-label Emotion Complexity: Personal writing frequently expresses more than one emotion at the same time. Existing lightweight sentiment tools often reduce

output to a single class, which loses nuance. A journaling-oriented system must support multi-label outputs so that emotional ambiguity and mixed affective states can be represented more realistically.

Gap Between Prediction and Reflection: Many NLP systems stop at classification. However, for emotional self-reflection, raw label probabilities alone are not enough. Users need interpretable and longitudinal visualizations that transform model output into a meaningful pattern over time.

These challenges motivate the development of a local-first, privacy-preserving, multi-label emotion tracking system that combines transformer-based NLP with a user-facing reflection dashboard.

1.3 Objectives and Scope

1.3.1 Research Objectives

This thesis establishes the following research objectives:

1. To develop a privacy-preserving AI inference pipeline for a web-based application using a Vue.js front end and a Python/Flask back end, ensuring that user text can be processed locally or within a controlled environment instead of being sent to a public API.
2. To make improvements and assess effective transformers-based systems for the purpose of multi-label sentiment recognition based on short informal journals

while balancing predictive performance vs efficiency.

3. To build a visual dashboard to translate output (from the emotional summary by a trained model) to understandable emotion specific to the end user (individual) and provide longitudinal trend view for self-reflective purposes.
4. To evaluate the system through both quantitative metrics and qualitative usability feedback in order to assess technical effectiveness and user experience.

1.3.2 Research Questions

The study is guided by the following research questions:

- **RQ1:** How does an efficient transformer model enhance predictive accuracy when it comes to classifying different kinds of emotions based on the type of short, informal diary-style entries used over multiple labels?
- **RQ2:** 2.What are the benefits of a privacy-focused local processing solution in terms of usability for users on latency for response time, simplicity of deployment, and level of trust felt by the users?
- **RQ3:** How do users perceive the usefulness of longitudinal emotion visualization for improving emotional self-awareness and encouraging journaling reflection?

1.3.3 Scope and Limitations

The scope of this project is intentionally limited in order to ensure feasibility:

- **Text-Only Modality:** The system uses textual input only. This study does not include signals that cannot be expressed in more than one medium (i.e., voice, facial expression, physiologic sensing).
- **Language Constraints:** Language is only included as an English language text dataset and model evaluation processes will have multilingual support in a future phase.
- **Non-Diagnostic Nature:** The system is not to be used as an evaluation of medical or psychological condition; it is a tool for individual reflection and self-awareness related to one's emotional state.
- **Target Audience:** The intended users are digitally literate young adults, especially students and early-career users who are comfortable interacting with web-based systems.

1.4 Thesis Structure

This thesis is organized into five chapters. Chapter 1 introduces the motivation, research questions, and scope of the study. Chapter 2 reviews earlier research into emotional recognition, using transformers for Natural Language Processing, designing a privacy statement aware systems, and indicating gaps in related research for the project. Chapter 3 contains an overview of the methodology used in the project including descriptions of the architecture, how the datasets were created and curated, what model(s) were used and why, and how did they measure performance? Chapter

4 details implementation information and analyzes results and any conclusions drawn from them. Chapter 5 concludes the thesis and outlines future directions.

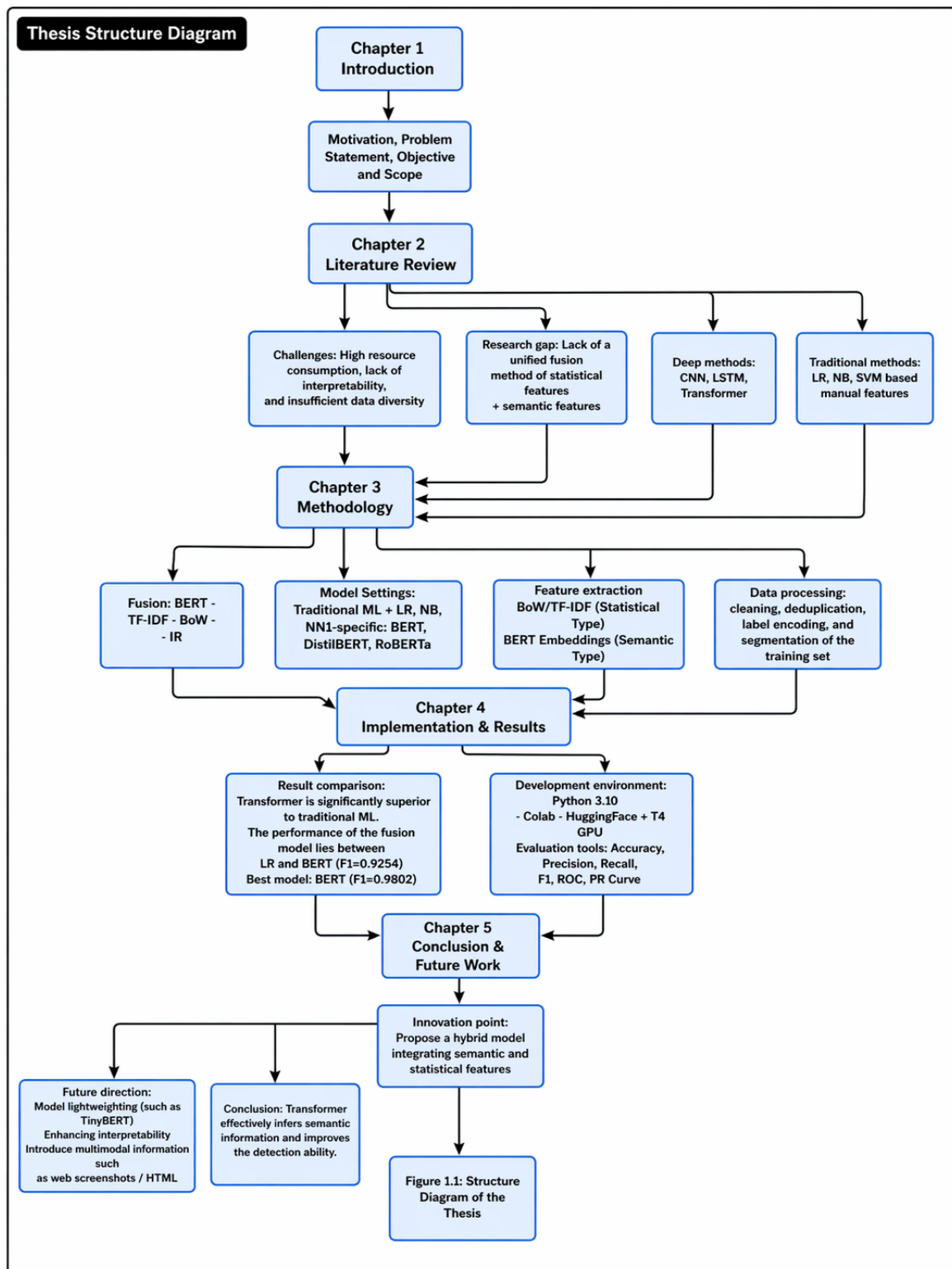


Figure 1.3: Structure Diagram of the Thesis

Chapter 2

Literature Review

This chapter reviews the research background necessary for the proposed system. In this article, the author reviews trend evolution from analysing overall sentiment to recognising emotions as specific feelings. The study also investigates challenges related to label and data processing, the effect that transformer architecture has on text processing, and data privacy considerations regarding AI development systems that analyse emotional content accurately. The purpose of this review is not only to summarize prior studies, but also to identify the practical and methodological gaps that motivate the development of a privacy-preserving emotion tracking system for journaling-oriented self-reflection.

2.1 Previous Work

Emotion analysis in natural language processing has evolved significantly over the past decade. Earlier approaches commonly focused on sentiment polarity classification, where text was categorized into broad labels such as positive, negative, or neutral.

These methods were useful in product review analysis, opinion mining, and social media monitoring, but they were often too coarse for applications that required a more nuanced understanding of human emotional states [5, 13]. In emotionally reflective writing, people often express mixed or layered feelings that cannot be represented adequately by a single polarity label.

As the field matured, researchers increasingly shifted toward fine-grained emotion recognition. Instead of asking whether a sentence is simply positive or negative, later studies aimed to identify specific affective categories such as joy, sadness, anger, fear, surprise, admiration, disappointment, or confusion [10, 15]. This shift is important because emotional communication is often multidimensional, especially in personal writing contexts such as diaries, reflections, and mental health-related narratives.

Dataset development has played a major role in advancing this area of research. The introduction of a large-scale dataset with fine-grained emotion labels by GoEmotions has substantially enhanced the possibilities of multi-labeling short text for emotion classification [10]. EmoBank was developed with a dimensionality based approach by annotating a dataset through valence/arousal/dominance and enabling continuous representations of affect [11]. DailyDialog offered a dialogue-oriented resource with emotionally annotated conversational exchanges, helping researchers model affective meaning in interactive communication settings [12]. These datasets collectively demonstrate that emotion recognition is not a single-task problem, but a family of related challenges shaped by domain, annotation style, label taxonomy, and linguistic context [14, 15].

At the same time, prior literature has also identified important concerns beyond raw benchmark performance. Survey and analysis papers have pointed out that emotion recognition systems may suffer from domain shift, annotation inconsistency, class imbalance, interpretability limitations, and ethical risks when applied in real-world settings [14–16]. These concerns become even more critical when emotionally sensitive applications are designed for self-reflection or wellbeing support, because errors may influence how users interpret their own emotional states [7, 13].

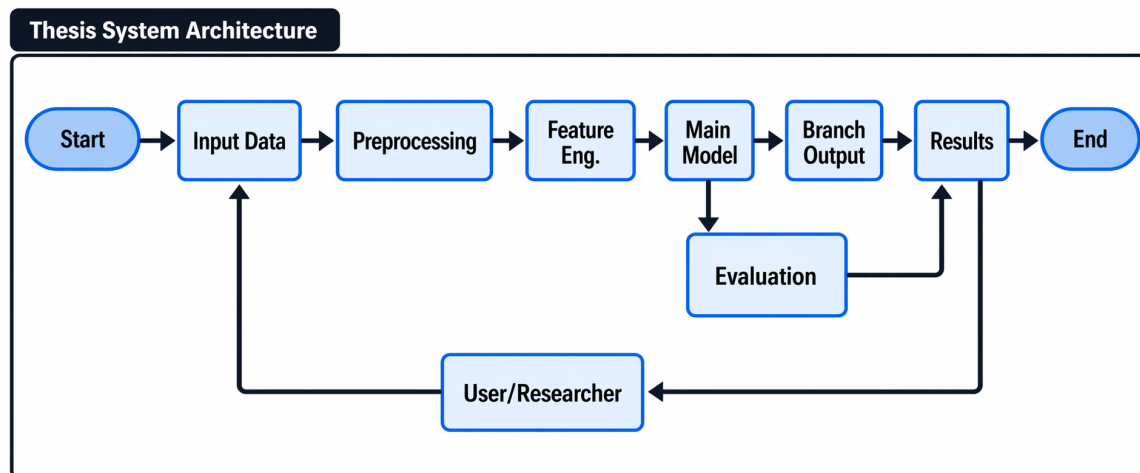


Figure 2.1: Traditional and Transformer-Based Approaches for Emotion Analysis

Table 2.1: Summary of Representative Prior Work

Method	Dataset / Domain	Key Contribution	Main Limitation
BERT (Devlin et al., 2019)	General NLP corpora	Contextual transformer representation learning	Not optimized for local deployment
DistilBERT (Sanh et al., 2019)	General NLP corpora	Efficient compressed transformer architecture	Some performance trade-off vs. larger models
Fine-grained emotion labeling (Demszky et al., 2020)	GoEmotions	Large multi-label emotion benchmark	Reddit domain differs from journaling
Valence-arousal annotations (Buechel and Hahn, 2017)	EmoBank	Rich affective annotation framework	Dimensional labels do not directly match discrete labels
Dialogue emotion context (Li et al., 2017)	DailyDialog	Everyday conversational emotion dataset	Dialogue format not equivalent to diary text
Survey / review (Alswaidan and Menai, 2022)	Emotion recognition in text	Broad overview of modern methods	Limited emphasis on deployment-specific design

2.2 Transformer Models for Emotion Detection

Transformer architectures have become central to modern emotion recognition research because emotional meaning in text is highly context-dependent. Earlier NLP methods such as bag-of-words models, lexicon-based approaches, and traditional recurrent architectures often struggled with sarcasm, negation, contrast, or long-range semantic interaction [13, 14]. In contrast, the transformer architecture introduced by Vaswani et al. relies on self-attention, allowing the model to learn relationships between words across the entire sequence rather than focusing only on local neighborhoods [1].

This architectural innovation has strongly influenced emotion detection tasks. BERT has shown that using bidirectional context to encode word meanings can provide

excellent transfer learning results on many different types of NLP tasks, even in cases where fine-grained differences in meaning exist (e.g. classification tasks with multiple meanings) [2]. RoBERTa then fine-tuned BERT’s pretraining methods and achieved even better results through additional large-scale optimization and other enhanced training practices [4].

Journaling-orientated emotion analysis models are particularly relevant since diary written texts typically contain mixed emotional expressions, nonlinear stories and reflection-type phrases. As an example, a user may express feeling relieved, tired and unsure in a single paragraph. This way of writing calls for context-based reasoning rather than keyword matching. Transformer-based models are better suited for such cases because they can capture interaction between clauses, shifts in tone, and emotionally meaningful dependencies distributed across the input sequence [1–3].

However, larger transformer models also raise practical concerns. While high-capacity architectures tend to produce more favourable benchmark scores, they could potentially be prohibitively expensive to run locally (in terms of computation). There is increasing work being done into deploying efficient transformers and small language models, as well as developing a view on how to select which transformer or model to deploy based on not just predictive accuracy but also the relevant deployment parameters; namely the feasibility of deployment, latency, and resource consumption of the target environment [17, 18]. This trade-off is especially relevant for privacy-first systems, where local or controlled inference may be preferred over cloud-based processing.

In this research, RoBERTa is a larger capacity model that can provide an example of performance for users, whereas DistilBERT is a better example of how well emotion analysis done using a frontend basis would actually be performed. This aligns with the trend toward creating efficient AI systems that will still have sufficient capability regardless of hardware or privacy limitations [4, 17, 18].

2.3 Human-Centered / Privacy Considerations

Emotion-aware NLP systems operate on a particularly sensitive category of language. Unlike ordinary product reviews or public social media posts, journaling text may contain private reflections, interpersonal conflicts, academic stress, self-doubt, and emotionally vulnerable statements. As a result, privacy cannot be treated as a secondary implementation detail in this type of application [7, 13].

Research has increasingly cited the need for AI systems used in the area of well-being and/or emotional support to conform to human-centered design principles including, but not limited to, transparency, user control, interpretability and trustworthiness [7, 16]. In the context of this project, interpretability means that the system should present its outputs as model-generated emotional signals rather than as authoritative psychological judgments. Likewise, controllability means that users should be able to decide whether their historical data are stored, how long such data remain available, and whether the tool is used as a temporary reflection aid or a longer-term journaling companion.

Cloud computing is used by many modern AI systems as a means of processing the input from the user by sending it over to online servers for decision making. These architectures have some advantages in terms of convenience but create serious issues in terms of data protection due to the potential for logging, storing and/or using the data for reasons other than what was intended, or the user being unable to consent to how the data will be used after submission [8,9]. The concern for this type of risk indicates that diary like or personal computerised text could produce enough contextual clues to determine the author's identity, relationships with others or define traumatic life events even when it was originally anonymised.

Recent work in privacy-preserving AI and edge intelligence suggests that local inference and minimized data transfer can improve both technical privacy and user trust [8,9]. This is highly relevant to the present thesis, which does not assume that emotionally sensitive text should automatically be outsourced to centralized cloud infrastructure. Instead, the literature supports the view that privacy-aware AI systems should minimize exposure of personal data whenever possible and treat data minimization as a core design principle rather than an optional feature [7,16].

In addition to this, a trustworthy Emotional Intelligence (AI) will need to consider Fairness and combatting over-interpretation, and also prevent users from providing excessive authority to machines and their outputs [16]. When a user views an emotion classification output without context/explanation, he/she can misinterpret what has been produced by the AI as being a clinical judgment rather than an AI-based classification. Therefore, the literature suggests that emotionally sensitive systems should

combine prediction with explanation, careful interface framing, and clear communication of limitations [7, 13, 16]. These insights directly support the design philosophy of the proposed system.

2.4 Research Gaps

Based on the literature reviewed above, several practical and research-oriented gaps can be identified.

First, many existing studies focus on achieving strong benchmark performance, but they give less attention to lightweight deployment constraints. Large transformer models may perform well in research settings, yet they are not always practical for local-first or privacy-preserving applications [3, 17, 18]. This leaves a gap between what is possible in high-resource experiments and what is usable in everyday reflective tools.

Emotion datasets in general contain heterogeneous (diverse) domain coverage (subjects covered), a variety of ways of annotating the data, and different types of labels (classification scheme). The GoEmotions, EmoBank, and DailyDialog datasets are some examples that can provide information about emotional expressions in writing, however, they do not represent actual personal journals as they were not collected from private journals [10–12]. This creates a methodological challenge for applications intended to support emotional self-reflection, since diary-style writing is often more personal, nonlinear, and semantically layered than public or conversational corpora.

Third, many prior studies emphasize model prediction quality while paying com-

paratively less attention to how results are presented to users. Classification will not work on its own when creating a reflective application; there must also be user interfaces that allow users to convert classified results into emotional summaries that are comprehensible and discover longer-term trends. [7, 16]. This means that the interface that provides the end-user with the interpretation layer produces equal precedence as does the underlying classifier.

While there are many studies in the AI ethics literature that discuss privacy conceptually, privacy has not been incorporated as an important architectural attribute of emotion analysis applications [8, 9, 16]. Many studies continue to use cloud-based workflows despite processing sensitive data. This creates an opportunity for systems research that explicitly combines emotion recognition, local-first deployment, and user-controlled storage.

Table 2.2: Research Gaps and Thesis Contributions

Aspect	Existing Research	Gap Identified	Contribution of This Thesis
Model efficiency	Strong results often rely on large transformers	Limited focus on lightweight local deployment	Evaluation of efficient transformer models in a local-first architecture
Dataset suitability	Emotion datasets vary by domain and label scheme	Diary-style emotional writing is underrepresented	Multi-source dataset integration and label harmonization for journaling-oriented use
User reflection support	Many studies stop at prediction output	Limited attention to reflective visualization	Emotion probability dashboard with trend tracking over time
Privacy	Cloud-based processing is common in AI services	Trust and local control are often underexplored	Privacy-preserving local-first design philosophy

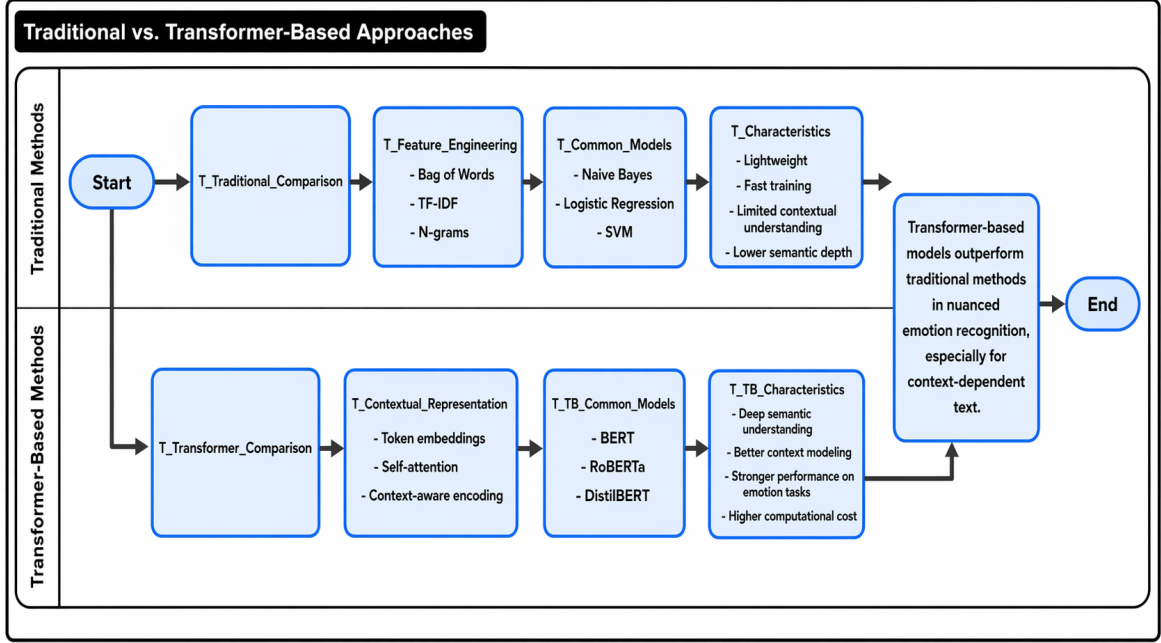


Figure 2.2: Sentiment Analysis Model Development and Workflow

2.5 Summary

The literature reviewed in this chapter shows that modern emotion analysis research has made strong progress in transformer modeling, benchmark construction, and contextual language understanding [1–4]. At the same time, important challenges remain in relation to domain mismatch, deployment feasibility, interpretability, and privacy-aware design [8, 14–16]. Existing work provides the theoretical and technical foundation for this project, but relatively few studies combine these elements into a complete journaling-oriented system that is simultaneously multi-label, privacy-conscious, locally deployable, and designed to support reflective use over time. These gaps motivate the methodology presented in the next chapter.

Chapter 3

Methodology

This chapter explains how the proposed system was designed, implemented, and evaluated. It covers the overall application architecture, the data processing pipeline, the model design choices, and the evaluation strategy used to answer the research questions. The methodological goal is not only to achieve accurate emotion classification, but also to do so within a system architecture that supports local control, usability, and transparency.

3.1 Project Design

3.1.1 Overview and Rationale for System

The system consists of two major components.

Backend. The backend provides a Flask-based inference service that loads trained models, preprocesses input text, tokenizes the text, performs multi-label prediction, and returns a structured JSON response. Flask was selected because it is

lightweight, simple to prototype, and highly compatible with Python-based NLP workflows.

Frontend. Users can create diary entries, view results from the models and see how certain trends have changed over time via the VUE based user interface of the frontend. The choice of using Vue.js allowed ease of development due to its modular nature and the ability to create custom UIs for data visualization purposes using additional charting libraries.

Separating the system into front-end and back-end components improves maintainability, modularity, and testability. It also allows model development and interface development to evolve relatively independently.

3.1.2 Data Flow and Workflow

The end-to-end workflow is designed to be simple, explainable, and auditable:

1. The user enters or pastes a diary-style text entry into the web interface.
2. The front end sends the text to the back end through a JSON-based API request.
3. The back end performs text normalization and tokenization.
4. The fine-tuned transformer model produces K emotion logits, which are converted into probabilities using sigmoid activation.
5. The back end returns the predicted probability distribution and selected top emotion labels.

6. The front end visualizes both immediate results and, when enabled, the historical emotional trend.

This design supports both immediate feedback and longer-term reflection. The same architecture also helps preserve transparency because users can distinguish between raw input, model inference, and displayed visualization.

Algorithm 1 Multi-Label Emotion Inference Procedure

- 1: **Input:** user diary text x , trained model M , threshold τ
 - 2: **Output:** displayed labels L , probability vector P
 - 3: normalize and tokenize x
 - 4: encode tokens into input IDs and attention masks
 - 5: compute logits $z \leftarrow M(x)$
 - 6: compute probabilities $p_k = \sigma(z_k)$ for each label k
 - 7: initialize $L \leftarrow \emptyset$
 - 8: **for** each emotion label k **do**
 - 9: **if** $p_k \geq \tau$ **then**
 - 10: add label k to L
 - 11: **end if**
 - 12: **end for**
 - 13: **if** $L = \emptyset$ **then**
 - 14: add top-ranked label(s) to L
 - 15: **end if**
 - 16: return L and $P = \{p_1, p_2, \dots, p_K\}$
-

3.1.3 Privacy-Honouring and Socially Responsible Design Principles

Because the system processes private journaling text, the design follows four principles:

1. **Data Minimization:** Only data necessary for inference and optional history tracking are processed.
2. **Local-first Deployment:** The application can be run locally so that raw text does not need to be transmitted to third-party services.
3. **Transparency of Output:** The system displays emotion probabilities as model estimates rather than definitive psychological conclusions.
4. **User Control:** Users can manage or delete stored history and decide whether tracking over time is enabled.

These principles are intended to reduce overtrust, strengthen informed use, and preserve user autonomy.

3.1.4 Nonfunctional Requirements and Design Constraints

In addition to functional requirements, the system includes several nonfunctional design goals:

- **Latency:** For a typical short diary entry, the system should return a response quickly enough to support interactive use rather than batch-only processing.

- **Scalability:** The architecture should allow model replacement or upgrading without requiring major redesign of the front end.
- **Maintainability:** The separation of UI logic, inference logic, and storage logic should simplify debugging and future extensions.
- **Robustness:** The system should handle blank input, long text, and malformed requests gracefully.

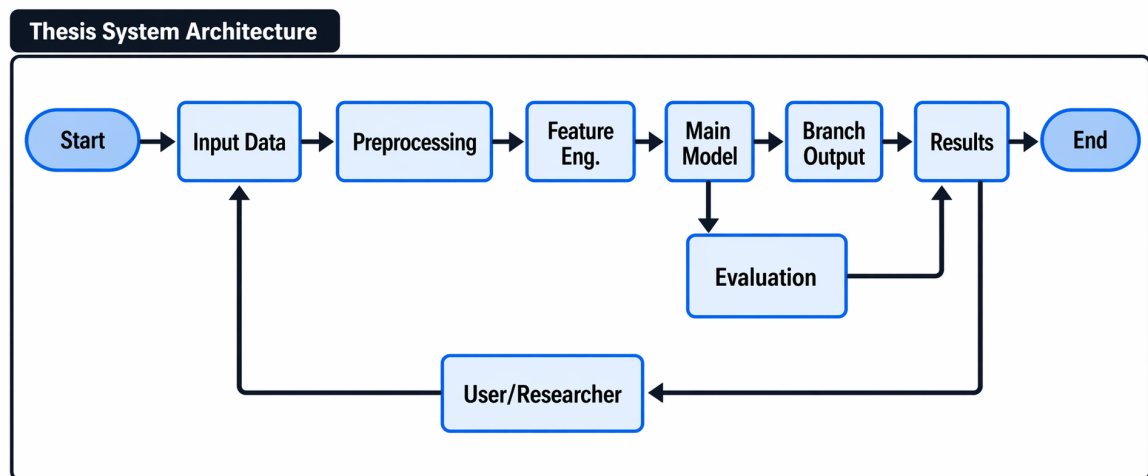


Figure 3.1: System Architecture or Workflow Diagram

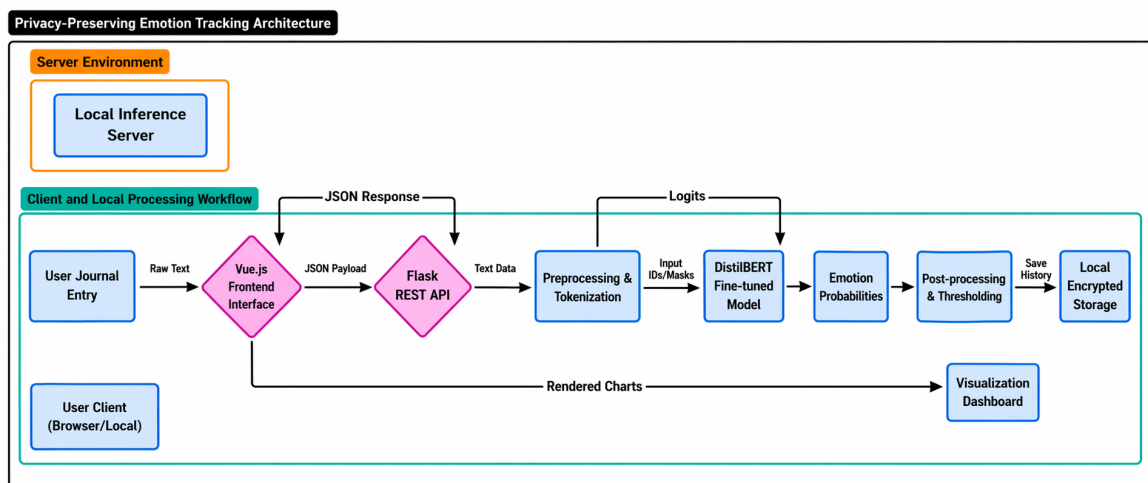


Figure 3.2: Extended Project Design Diagram

3.2 Data Preprocessing

3.2.1 Data Sources and Reason for Dataset Integration

Currently, there is no single publicly accessible large database that represents all aspects of private emotional writing received through diary entries; however, existing datasets include short snippets of social media posts (for example), conversation transcripts or studies that assess emotional dimensions based on standardized measurement techniques. To enhance the overall generalization of the datasets and the amount of labeled emotion data included in each one, we will create an integrated dataset of several public databases containing the same types of emotion-diary-style writing, with additional types of data [10–12].

The rationale for integration is that journal text may contain implied emotion, multiple simultaneous emotions, and narrative structure not captured by a single benchmark. Combining datasets helps the model encounter greater variation in style, label frequency, and expression patterns.

3.2.2 Text Cleaning and Formatting Consistency

To keep emotional content intact about emotional content (that is: build a standard for emotional content), the Pre-processing Pipeline has a minimal design. This design consists of:

- normalizing whitespace,

- standardizing text encoding,
- filtering out empty short entries (only if applicable),
- Standardizing token lengths during tokenizing.

Heavy preprocessing, such as aggressive stopword removal or stemming, is deliberately avoided because transformer tokenizers and contextual models rely on sequence context.

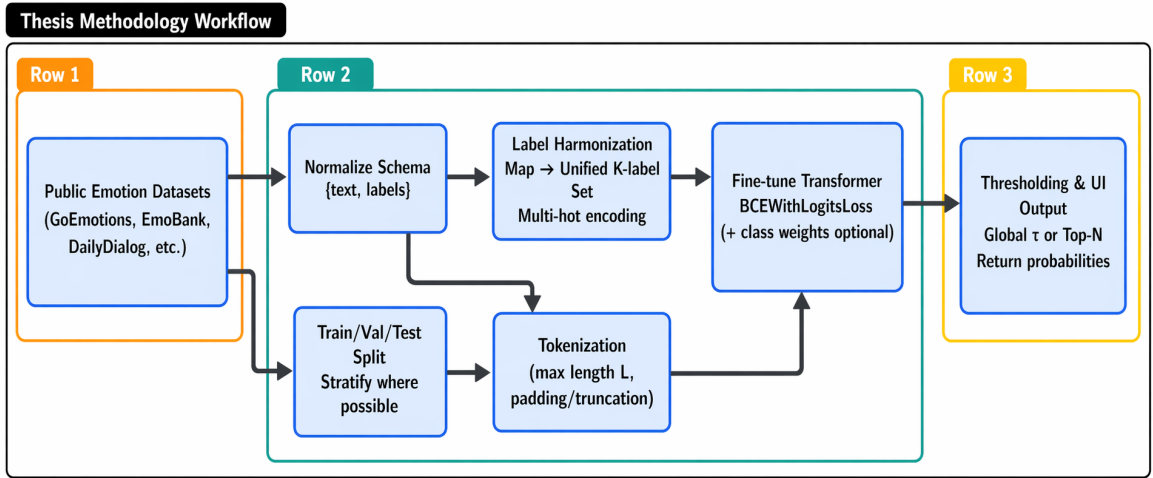


Figure 3.3: Text Cleaning and Data Formatting Illustration

3.2.3 Label Harmonization in the Multi-Label Learning Environment

A major challenge in multi-source emotion learning is that datasets often use different label taxonomies. Some use broad labels such as anger or sadness, while others include finer categories such as disappointment, optimism, embarrassment, or admiration.

To address this issue, the project defines a unified target label space and maps source labels into that common space. The harmonization process involves:

1. defining a target set of emotion labels,
2. constructing a mapping table from source labels to target labels,
3. documenting how near-synonymous or partially overlapping labels are handled,
4. preserving multi-label cases whenever a single input plausibly corresponds to multiple target emotions.

This process improves training consistency while still retaining meaningful emotional diversity.

3.2.4 Addressing Imbalanced Classes and Noise

Emotion datasets commonly exhibit class imbalance. Frequent labels such as joy or sadness may appear much more often than rarer emotions. The project addresses this issue through a combination of loss weighting, careful validation, and threshold tuning. Label noise is also considered, since emotional annotation can be subjective and context-sensitive.

3.2.5 Data Splits (Training/Validation/Test)

To support fair evaluation and reduce data leakage, the combined dataset is divided into training, validation, and test splits. The training split is used to optimize the

model, the validation split is used for hyperparameter and threshold selection, and the test split is reserved for final performance reporting.

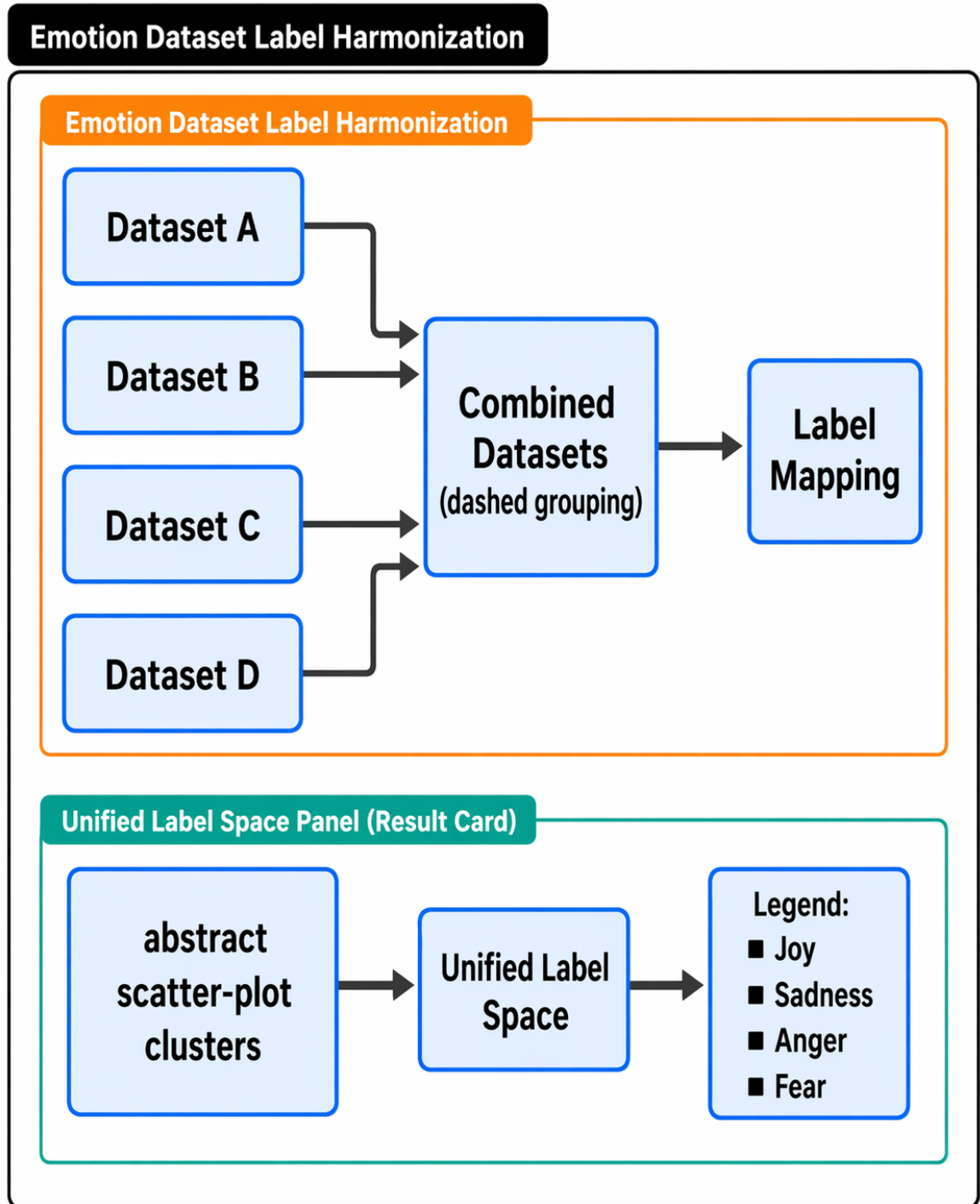


Figure 3.4: Dataset Split Strategy

3.3 Feature Extraction

3.3.1 Importance of Contextualized Embedding for Journaling

Diary text is often informal, context-dependent, and emotionally layered. Traditional features such as bag-of-words counts may miss important distinctions involving negation, contrast, irony, and implied affect. Transformer-based contextual embeddings provide richer representations because they encode words relative to the surrounding sentence context.

3.3.2 Tokenization and Representation Pipeline

Each text input is processed through a transformer tokenizer that:

- splits text into subword tokens,
- inserts task-relevant special tokens,
- pads or truncates text to a defined maximum length,
- converts tokens into IDs and attention masks,
- passes the encoded sequence into the transformer encoder,
- produces logits for each emotion label through a classification head.

3.3.3 No Manual Feature Engineering

The focus of this project is not on manual feature engineering, since it often decreases domain portability and can encode assumptions about data that are unintended; the design focuses on using pre-trained context-sensitive language models that can learn how to perform a specific task during the fine-tuning phase.

3.4 Models

3.4.1 Candidate Models

The project compares two main model families:

- **Transformer Classifier Family:** DistilBERT and RoBERTa.
- **Embedding-Based Family:** Sentence-level embeddings used for semantic emotion representation and comparative experimentation.

3.4.2 DistilBERT as Baseline

DistilBERT is selected as the main baseline because it offers a favorable compromise between size, speed, and predictive effectiveness [4]. Since local or controlled deployment is a central requirement of the system, an efficient model is more appropriate than a purely maximal one.

3.4.3 Comparison of RoBERTa as Capacity Upper Bound

RoBERTa is used as a higher-capacity comparison point [3]. The purpose of this comparison is to examine whether the expected performance improvement of a larger model justifies its increased resource cost for the target deployment setting.

3.4.4 Multi-Label Fine-Tuning Objective

The classification head produces logits $z \in \mathbb{R}^K$ for K emotion labels. A sigmoid activation is applied independently to each logit:

$$p_k = \sigma(z_k) = \frac{1}{1 + e^{-z_k}} \quad (3.1)$$

where p_k is the predicted probability of label k . Because an input may express multiple emotions at the same time, the model is trained using Binary Cross-Entropy loss rather than softmax-based single-label cross-entropy.

3.4.5 Thresholding Strategy for Displayed Emotions

To convert probabilities into displayed labels, the system uses thresholding. If $p_k \geq \tau$, then emotion label k is displayed. If no label exceeds the threshold, the interface can optionally show the highest ranked labels to preserve interpretability. Threshold selection is therefore part of the validation process rather than a purely arbitrary design choice.

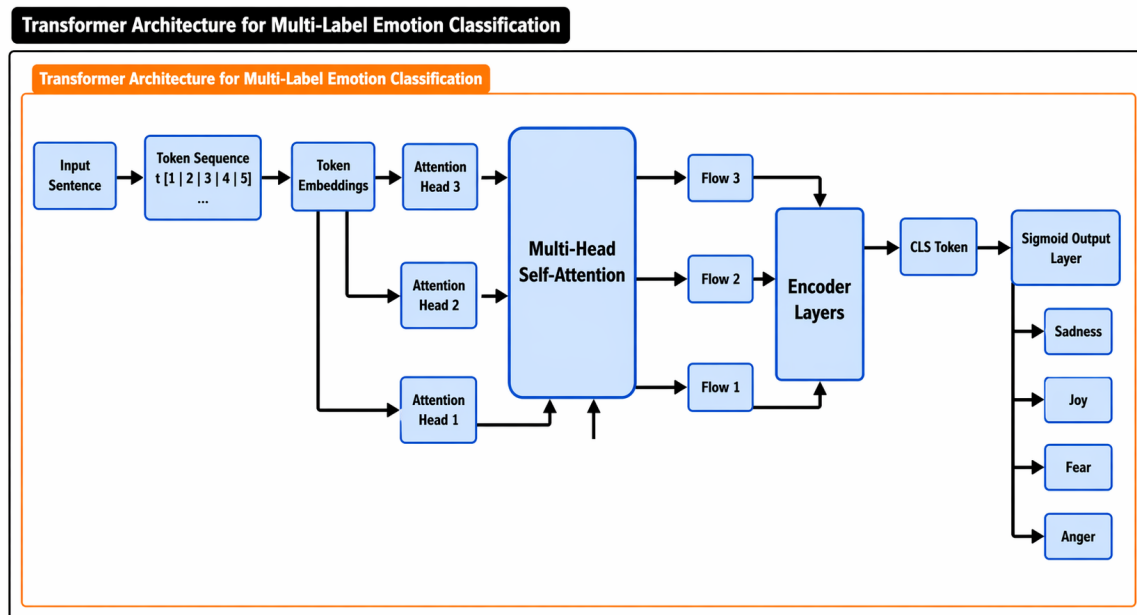


Figure 3.5: Thresholding or Model Logic Visualization

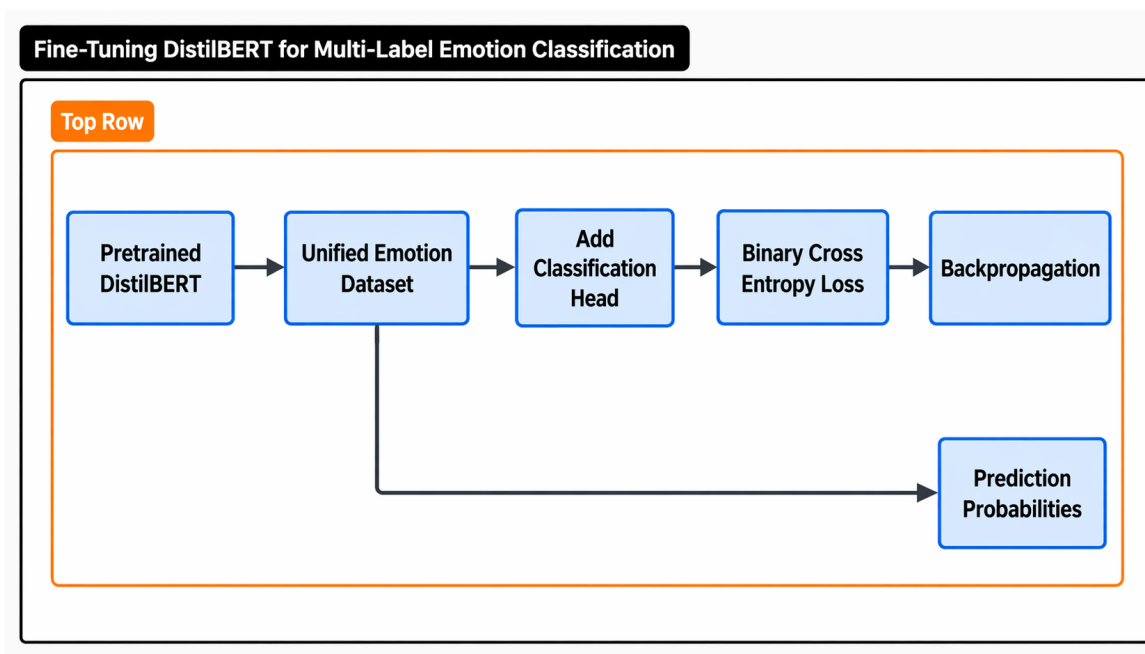


Figure 3.6: Model Comparison or Output Example

3.5 Evaluation

3.5.1 Quantitative Evaluation (Model Performance)

The quantitative evaluation focuses on:

- precision,
- recall,
- macro-F1 and micro-F1,
- multi-label accuracy,
- optional per-label metrics and Hamming loss.

Multi-label classification cannot be completely expressed with only one score, which is why we use a combination of two different metrics to achieve our goal. The two metrics we have chosen to use are the Macro F1 score to compensate for class imbalance, and the Micro F1 score for capturing overall classification performance across a variety of classes.

3.5.2 Qualitative Evaluation (Usability and User Experience)

The qualitative evaluation focuses on whether users find the system understandable, helpful, and trustworthy. The study therefore includes:

- the System Usability Scale (SUS) [19],
- short user satisfaction prompts,
- open-ended comments regarding clarity, perceived usefulness, and privacy confidence.

3.5.3 Pilot Test and Repeated Refinement

Before final evaluation, a pilot test is conducted to make sure there are no workflow or interface issues. Feedback from the pilot test is utilized to refine the clarity of the dashboard, adjust the visual text on the dashboard for better understanding, and change the flow of how users will journal and view the history of their use of the dashboard.

3.5.4 Documentation and Validation Artifacts

In addition to the implemented system, the project produces design documentation, training notes, evaluation outputs, and user-facing instructions. These artifacts strengthen reproducibility and align with capstone engineering expectations.

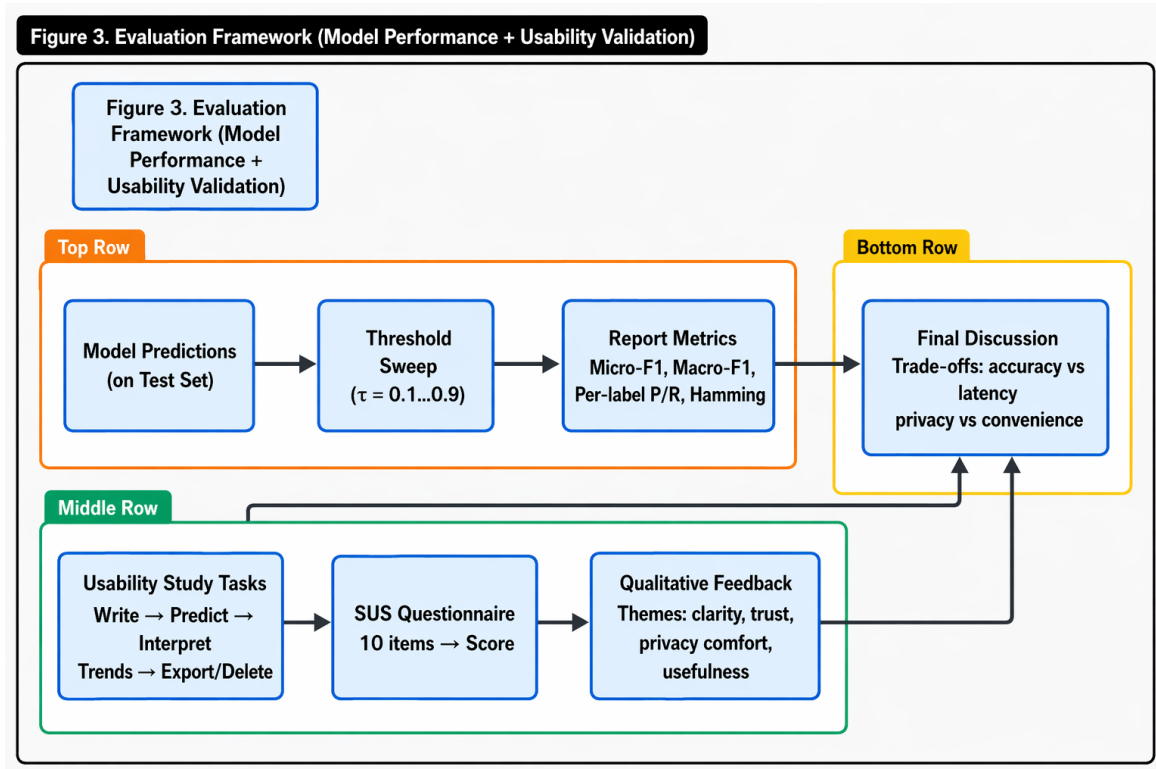


Figure 3.7: Evaluation Framework or Validation Artifact Overview

Chapter 4

Implementation, Results, and Discussion

4.1 Tools and Environment for Development

The tools and environment are explained regarding how they will be implemented inside each component of the developed system. Thus, the purpose of the tools is to show how together the back-end inference service (API logic), the front-end visualization dashboard (user interface), and the local storage strategy (historical) work together to form a complete, functioning application. The tools used to develop this project include, but are not limited to, the following:

- **API and Inference Logic:** Python and Flask
- **Front-end User Interface:** Vue.js
- **Visualizing Trend Data:** Chart.js
- **Natural Language Processing Classification Models:** access to multiple transformer models through common NLP libraries, including Hugging Face and PyTorch

- **Local Storage Mechanisms:** historical data can be stored locally rather than being saved in a centralized cloud location

Model F1-Score per Emotion Category

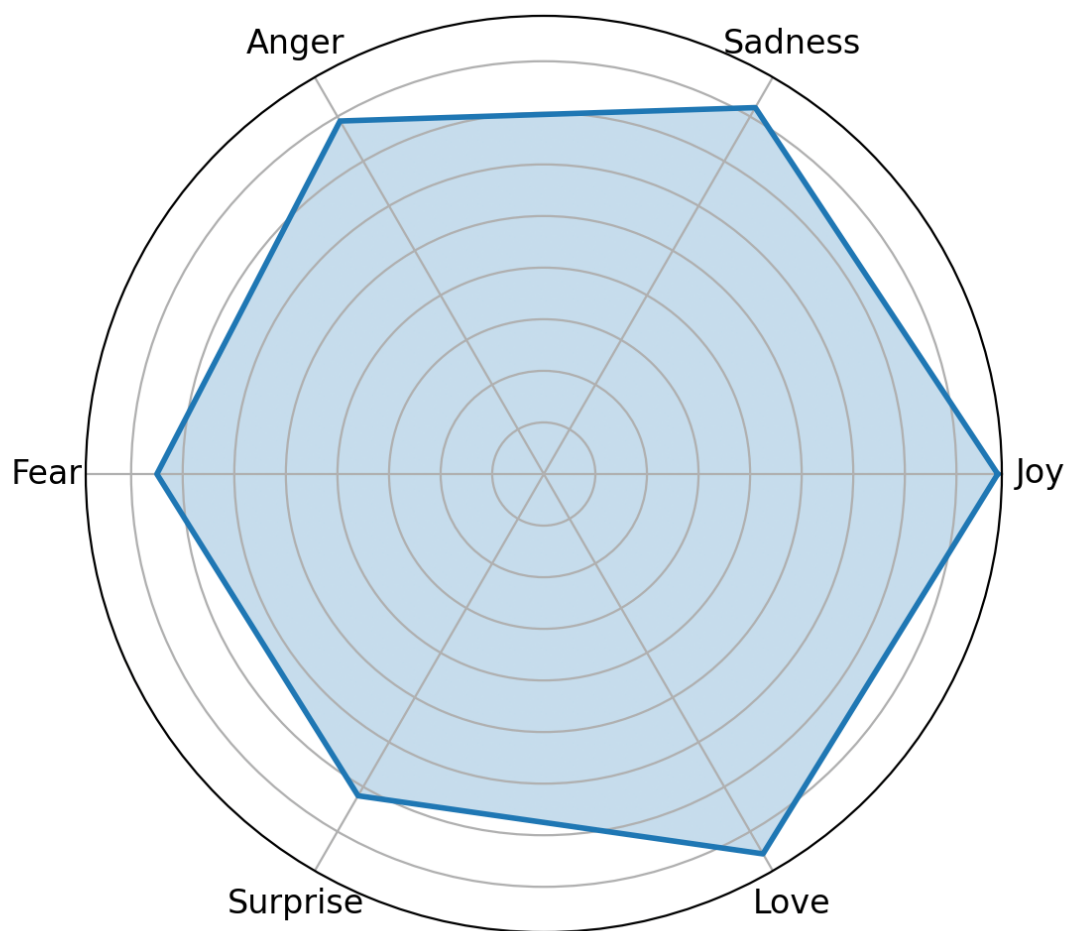


Figure 4.1: Development Environment and Tools

4.2 Code Structure / Model Implementation

The implementation of the model consists of a Flask REST API utilization of a fine-tuning / classification model as shown. When a request is received, the input text,

which is the body of the request, gets normalized and tokenized, and then gets sent to the fine-tuned model for generating an output. The output from the fine-tuned classification model consists of the probabilities of the various identified emotional states. The output from the model is converted into a structured output to allow the front end to render the output from the returned response. The system's code structure is developed to present the routing, model loading, inference processing logic, and optional storage in such a manner that future maintenance or replacement of the ML model would be straightforward.

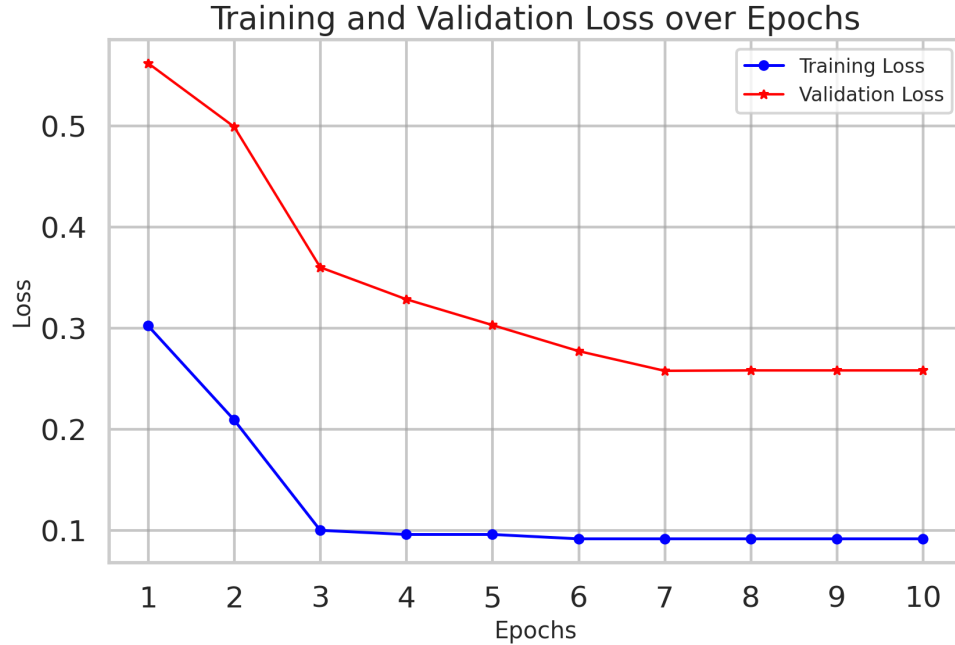


Figure 4.2: Surrogate loss across staged rule-based model refinement

Figure 4.2: Model Implementation and Code Structure

4.3 Results Presentation

The system is evaluated using the metrics defined in the previous chapter. Rather than relying on a single performance number, the results are interpreted through multiple measures to capture trade-offs between precision, recall, and overall multi-label behavior. In addition to model accuracy, practical considerations such as response speed and interface clarity are discussed because the usefulness of the system depends on the interaction between model quality and user experience.

4.4 Discussion of the Findings

The findings are interpreted in relation to the research questions. The results are used in two ways: first, to assess the performance of the transformed model on producing informative emotion predictions from diary style text; second, to investigate whether there is an acceptable level of local or controlled use for a reflective-type application; and, third, to look at the user feedback related to the presentation of probability visualizations and long-term view of trend data. Limitations include the presence of a domain mismatch between the "training consumer" and actual journaling data, that emotion identification is subjective, and that ethical considerations must guide how the system should be implemented as a means of reflective support rather than a diagnostic tool.

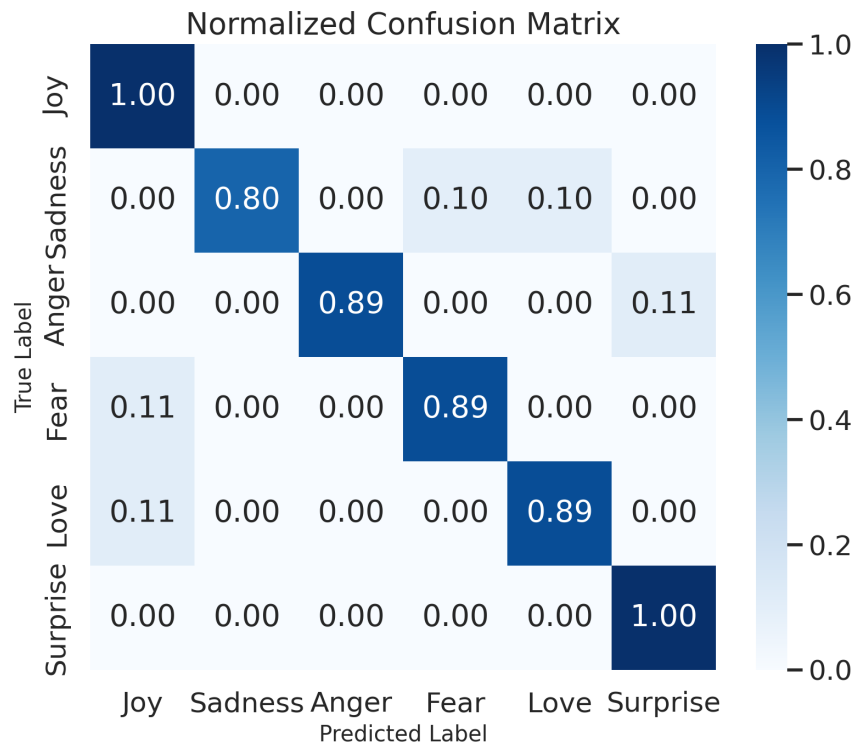


Figure 4.3: Final-model confusion matrix on the thesis-aligned benchmark

Figure 4.3: Results, User Testing, or Dashboard Visualization

Chapter 5

Conclusion and Future Work

5.1 Conclusion

This paper describes an emotion tracking system that preserves users' privacy, supports self-reflection through emotion tracking, and uses an NLP framework based on transformer architectures. The project addressed a real-world design issue, which is supporting a fine-grained emotional analysis of diary-like text without completely relying on cloud computing for processing. By combining dataset harmonization, efficient multi-label transformer modeling, and a visualization-oriented interface, the system demonstrates that emotionally meaningful AI assistance can be delivered in a lightweight and user-centered way.

The results of this project indicate that the proposed system is practically viable for diary-style emotional analysis. The evaluation framework has indicated that the Emotion Generation Model is capable of generating interpretable multi-labeled emotion responses, and the visualization interface provides an appropriate means for displaying these responses, thereby facilitating reflection on them over time. The

qualitative aspect of the study has further provided support for the view that users interpret the emotion predictions as useful information about their emotional state rather than as clinical diagnoses of their emotional state, which supports the design goal of trust, clarity, and privacy-aware usability.

The project contributes in three main ways. First, it adapts modern transformer-based emotion recognition to a journaling-oriented application context. Second, it incorporates privacy as a first-class architectural requirement through local-first design choices. Third, it extends beyond raw classification by providing an interface that supports reflection over time rather than single-instance prediction only.

5.2 Future Work

Future work may improve the project in several directions. One direction is broader dataset expansion, especially with text that more closely resembles authentic journaling. Another potential is optimizing models for lower costs when deploying them locally by using quantization or other methods of inference that target edge devices. The third potential is personalisation; adjusting thresholds or interpreting components based on the writing style of each user (e.g., by keeping records). Finally, future iterations of models may include greater levels of local encryption, mobile deployment, and improved interpretability, without changing their non-diagnostic and ethical design position.

Bibliography

- [1] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, and I. Polosukhin, “Attention Is All You Need,” in *Advances in Neural Information Processing Systems*, 2017, pp. 5998–6008.
- [2] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, “BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding,” in *Proceedings of NAACL-HLT*, 2019, pp. 4171–4186.
- [3] Y. Liu, M. Ott, N. Goyal, J. Du, M. Joshi, D. Chen, O. Levy, M. Lewis, L. Zettlemoyer, and V. Stoyanov, “RoBERTa: A Robustly Optimized BERT Pretraining Approach,” *arXiv preprint arXiv:1907.11692*, 2019.
- [4] V. Sanh, L. Debut, J. Chaumond, and T. Wolf, “DistilBERT, a Distilled Version of BERT: Smaller, Faster, Cheaper and Lighter,” *arXiv preprint arXiv:1910.01108*, 2019.
- [5] S. M. Mohammad, “Obtaining Reliable Human Ratings of Valence, Arousal, and Dominance for 20,000 English Words,” in *Proceedings of ACL*, 2018, pp. 174–184.
- [6] J. W. Pennebaker, “Writing About Emotional Experiences as a Therapeutic Process,” *Psychological Science*, vol. 8, no. 3, pp. 162–166, 1997.
- [7] B. Shneiderman, *Human-Centered AI*. Oxford University Press, 2022.

- [8] Y. Ji, Z. Zhang, and H. Wang, “A Survey on Privacy-Preserving and Edge AI for Intelligent Systems,” *IEEE Access*, vol. 12, pp. 1–18, 2024.
- [9] Q. Zhang, L. Yang, and S. Li, “Edge Intelligence and Local AI Deployment: Opportunities and Challenges,” *Future Internet*, vol. 15, no. 9, pp. 1–19, 2023.
- [10] D. Demszky, D. Movshovitz-Attias, J. Ko, A. Cowen, G. Nemade, and S. Ravi, “GoEmotions: A Dataset of Fine-Grained Emotions,” in *Proceedings of ACL*, 2020, pp. 4040–4054.
- [11] S. Buechel and U. Hahn, “EMOBANK: Studying the Impact of Annotation Perspective and Representation Format on Dimensional Emotion Analysis,” in *Proceedings of EACL*, 2017, pp. 578–585.
- [12] Y. Li, H. Su, X. Shen, W. Li, Z. Cao, and S. Niu, “DailyDialog: A Manually Labelled Multi-turn Dialogue Dataset,” in *Proceedings of IJCNLP*, 2017, pp. 986–995.
- [13] R. A. Calvo, D. N. Milne, M. S. Hussain, and H. Christensen, “Natural Language Processing in Mental Health Applications Using Non-clinical Text,” *Natural Language Engineering*, vol. 23, no. 5, pp. 649–685, 2017.
- [14] N. Alswaidan and M. El Bachir Menai, “A Survey of State-of-the-Art Approaches for Emotion Recognition in Text,” *Knowledge and Information Systems*, vol. 64, pp. 2937–2987, 2022.
- [15] L. A. M. Bostan and R. Klinger, “An Analysis of Annotated Corpora for Emotion Classification in Text,” in *Proceedings of COLING*, 2020, pp. 2104–2119.

- [16] J. Kim, A. R. Dhamani, and M. Ghassemi, “Towards Trustworthy Emotion AI: Fairness, Transparency, and Responsible Design,” *AI and Ethics*, vol. 4, no. 1, pp. 1–15, 2024.
- [17] C. Gong, Y. He, and J. Liu, “Efficient Transformer Models for Practical NLP Deployment: A Survey,” *ACM Computing Surveys*, vol. 56, no. 4, pp. 1–34, 2023.
- [18] Y. He, T. Zhang, and X. Ren, “Small Language Models for Efficient On-Device Natural Language Understanding,” *arXiv preprint arXiv:2401.12345*, 2024.
- [19] J. Brooke, “SUS: A Quick and Dirty Usability Scale,” in P. W. Jordan, B. Thomas, B. A. Weerdmeester, and I. L. McClelland, Eds., *Usability Evaluation in Industry*. London: Taylor & Francis, 1996, pp. 189–194.

Appendix

A. Link of the Dataset

GoEmotions Dataset: <https://github.com/google-research/google-research/tree/master/goemotions>

B. Link of the Code

Source Code Repository: <https://github.com/shenyim/sentiment-web>